Muhammad Saif Shakil

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Muhammadsaifff

PROFESSIONAL SUMMARY

Computer Science graduate from UIT University with hands-on experience in Artificial Intelligence, Digital IC Design & Verification, Mobile Application Development, Full-Stack Development, and Computer Vision. Skilled in Verilog, SystemVerilog, FPGA-based digital design, Flutter, Python, Node.js, Firebase, REST APIs, PostgreSQL, Oracle Database, and OpenCV through academic, research, internship, and industry-focused training projects. Passionate about designing efficient hardware systems, building practical software solutions, and continuously learning emerging technologies.

EXPERIENCE

Digital IC Design & Verification Trainee

(Ongoing) | INSPIRE - PSEB

- Selected for a fully funded national semiconductor training program.
- Learning RTL Design, verification methodologies, and semiconductor fundamentals.
- Selected for a fully funded national semiconductor upskilling program under the Prime Minister's Initiative.
- Gaining hands-on experience in RTL Design, Digital Logic Design, Verilog, SystemVerilog, FPGA-based design flow, and functional verification.
- Learning semiconductor fundamentals, simulation, synthesis, testbench development, and industry-standard EDA tools through practical laboratory sessions and projects.
- Participating in industry-oriented training conducted by NED University in collaboration with PSEB and Tech Destination Pakistan.

PROJECT

UART (Universal Asynchronous Receiver/Transmitter) | Verilog, FPGA, Digital Design

Designed and implemented a UART communication module in Verilog for FPGA-based asynchronous serial communication, including TX/RX modules, baud rate generation, start/stop bit detection, and serial/parallel data conversion. Verified functionality using Verilog testbenches and successfully validated the design on an FPGA development board.
Github Repo: https://github.com/Muhammadsaifff/uart

AMBA APB Master-Slave Interface | Verilog, FPGA, Digital Design

Designed and implemented AMBA APB Master and Slave modules in Verilog, supporting compliant read/write transactions with APB protocol signals. Developed FSM-based control logic, verified functionality using Verilog testbenches, and validated the design through simulation and FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/amba_apb

AXI4-Lite Communication Interface | Verilog, FPGA, Digital Design

Designed and implemented an AXI4-Lite Master-Slave interface in Verilog with the five AXI4-Lite channels and VALID/READY handshake protocol. Developed FSM-based control logic, verified functionality using Verilog testbenches, and validated the design through simulation and FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/axi-lite

AMBA AHB-Lite Master-Slave Interface | Verilog, FPGA, Digital Design

Designed and implemented an AMBA AHB-Lite Master and Slave interface in Verilog, supporting protocol-compliant read/write transactions with AHB-Lite signals. Developed FSM-based control logic, verified functionality using Verilog testbenches, and validated the design through simulation and FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/ahb

VGA Controller with SRAM Frame Buffer | Verilog, FPGA, Digital Design

Designed and implemented a VGA controller in Verilog with an SRAM-based frame buffer for real-time video output. Implemented VGA timing, synchronization, pixel addressing, and display control, verified functionality using Verilog testbenches, and validated the design through FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/vga_controller

Cache Controller Design & Verification | Verilog, FPGA, Digital Design

Designed and verified a direct-mapped cache controller in Verilog using SRAM-based cache and main memory. Implemented cache read/write operations, hit/miss detection, tag comparison, and FSM-based control logic. Verified functionality using Verilog testbenches and validated through simulation and FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/cache_controller

32-Bit Arithmetic Logic Unit (ALU) Design | Verilog, FPGA, Digital Design

Designed and implemented a 32-bit ALU in Verilog supporting arithmetic, logical, comparison, and shift operations. Developed combinational RTL, verified functionality using Verilog testbenches, and validated the design through simulation and FPGA implementation.
Github Repo: https://github.com/Muhammadsaifff/32bit_alu

EDUCATION

Bachelor of Science in Computer Science (CGPA: 3.52/4.00)

UIT University - (2022 - 2026)

SKILLS

Hardware Description Languages: Verilog, SystemVerilog

Digital Design & Computer Architecture: RTL Design, Combinational & Sequential Logic, Finite State Machines (FSMs), Clock Domain Crossing (CDC), SRAM-Based Memory Systems

AMBA & Communication Protocols: AMBA AXI4-Lite, AMBA APB, AMBA AHB, UART

FPGA & Verification: FPGA Design & Implementation, Functional Verification, Testbench Development, Simulation, Waveform Debugging

Programming Languages: Python, C++, Java, Dart, SQL, JavaScript

Mobile & Full-Stack Development: Flutter, FastAPI, Flask, Firebase, REST APIs

Databases: PostgreSQL, Oracle Database, Firebase Firestore, SQLite

AI & Machine Learning: Artificial Intelligence, Natural Language Processing (NLP), Prompt Engineering, OpenCV, Hugging Face Transformers, spaCy, Scikit-learn, Pandas, NumPy

Tools & Platforms: Git, GitHub, Docker, Android Studio, VS Code, Arduino IDE, Cisco Packet Tracer, Logisim, MATLAB, Power BI